

1. **In Pandas, what data structure is used to represent a one-dimensional labeled array?** (a) DataFrame (b) List (c) Array (d) Series
2. **In pandas, which method is used for creating a heatmap to visualize the relationship?** (a) distplot (b) barplot (c) joinplot (d) heatmapplot
3. **Polymorphism allows objects of different classes to be treated as objects of a:** (a) Class variable (b) Subclass (c) Superclass (d) Base class
4. **What is an inner or nested function in Python?** (a) A function with a global scope (b) A function that takes no arguments (c) A function defined inside another function (d) A function with no return statement
5. **What is the primary purpose of Anaconda in the context of AI development?** (a) Operating system installation (b) Text editing (c) Package management (d) Web development
6. **What is the primary purpose of data masking?** (a) To hide sensitive information (b) To encrypt data at rest (c) To implement access control lists (ACLs) (d) To implement role-based access control (RBAC)
7. **What is the primary purpose of integrating Jupyter or Python notebooks into a data architecture?** (a) Data visualization (b) Real-time data processing (c) Stream processing (d) Collaboration and documentation
8. **What is the primary purpose of the data exploration layer in a data architecture?** (a) To store raw data (b) To ensure data security (c) To provide a user-friendly interface for data visualization (d) To optimize data storage costs
9. **What is the purpose of a constructor in a Python class?** (a) To define inner functions (b) To destroy an object of the class (c) To perform operations on class variables (d) To create an object of the class
10. **What is the purpose of data cleansing in data transformation processes?** (a) To handle missing data (b) To ensure data accuracy and quality (c) To split data (d) To handle duplicate data
11. **What is the purpose of the reshape method in Numpy?** (a) Sort the elements of an array (b) Compute the standard deviation of an array (c) Calculate the mean of an array (d) Change the shape of an array
12. **What is the scope of a local variable in Python?** (a) Module-level (b) Function-level (local) (c) Global (d) Class-level
13. **Which data type is used to represent a single character in Python?** (a) Float (b) Character (c) String (d) Integer

14. **Which keyword is used to handle exceptions in Python?** (a) except (b) throw (c) try (d) catch
15. **Which loop in Python is used for iterating over a sequence?** (a) for loop (b) enumerate loop (c) nested loop (d) while loop
16. **Which of the following is an example of a continuous probability distribution?** (a) Bernoulli distribution (b) Poisson distribution (c) Normal distribution (d) Binomial distribution
17. **Which of the following is used to update and upgrade packages on a Debian-based system?** (a) apt-get update && apt-get upgrade (b) update (c) dpkg (d) apt-get install
18. **Which of the following statements defines a lambda function in Python?** (a) `def(x): x * 2` (b) `lambda x: x * 2` (c) `def double(x): return x * 2` (d) `lambda double(x): x * 2`
19. **Which type of plot displays the relationship between two variables?** (a) Bar chart (b) Histogram (c) Pie chart (d) Scatter plot
20. **The coefficient of variation is a measure of relative variability and is calculated as:** (a) Mode to standard deviation (b) Standard deviation to mean (c) Mean to median (d) Range to variance
21. **What does the "if" statement in Python allow you to do?** (a) Perform conditional execution of code (b) Repeat a block of code a specified number of times (c) Define a function (d) Declare a variable
22. **What does the "sudo" command allow you to do in a Unix-based system?** (a) Delete files (b) List files (c) Run commands with administrative privileges (d) Create directories
23. **What is the process of converting continuous data into discrete intervals or bins?** (a) Grouping (b) Binning (c) Discretization (d) Aggregation
24. **Which access specifier restricts access to class members only within the class?** (a) Private (b) Protected (c) Global (d) Public
25. **Which data structure in Python is ordered and mutable?** (a) List (b) Dictionary (c) Set (d) Tuple
26. **Which function is used to apply a given function to all items in an iterable?** (a) `reduce()` (b) `apply()` (c) `map()` (d) `filter()`
27. **Which library in Python is commonly used for numerical operations and array processing?** (a) Scikit-learn (b) Numpy (c) Matplotlib (d) Pandas

28. **Which method is used to combine or concatenate two Numpy arrays?** (a) join() (b) combine() (c) merge() (d) concatenate()
29. **Which of the following is a valid reason to implement a partition strategy for files?** (a) To decrease data query performance (b) To improve query performance and reduce query execution time (c) To simplify data organization (d) To increase data storage costs
30. **Which of the following is an example of qualitative data?** (a) Age in years (b) Gender (Male/Female) (c) Weight in kilograms (d) Temperature in degrees Celsius
31. **How do you select a subset of rows and columns from a DataFrame in Pandas?** (a) Using the filter() method (b) Using the select() method (c) Using indexing and slicing (d) Using the subset() method
32. **In pandas, which method allows you to group data based on one or more variables?** (a) Concatenate (b) Merge (c) Join (d) GroupBy
33. **What is the purpose of the "break" statement in a loop in Python?** (a) It checks a condition and exits the loop if it's true. (b) It skips the current iteration and continues with the next one. (c) It executes the loop code repeatedly. (d) It terminates the loop immediately.
34. **What measure of central tendency is most affected by outliers in a dataset?** (a) Mean (b) Standard deviation (c) Mode (d) Median
35. **Which of the following is NOT a common plot type in pandas for data visualization?** (a) Box plot (b) Radar plot (c) Histogram (d) Scatter plot
36. **Inheritance is a concept that allows a class to:** (a) Have multiple constructors (b) Access private class variables (c) Inherit properties and methods from another class (d) Create multiple instances of itself
37. **What is the purpose of an alias in a Unix-based system?** (a) It deletes files. (b) It creates a new user account. (c) It renames a directory. (d) It provides a shortcut for executing commands or command sequences.
38. **What type of operation in pandas is used to combine DataFrames with the same schema?** (a) Joining (b) GroupBy (c) Concatenation (d) Merging
39. **Which method is used to create a Numpy array from a Python list?** (a) numpy.array() (b) numpy.create_array() (c) numpy.arr() (d) numpy.from_list()
40. **Which command is used to list the files and directories in the current directory in Linux?** (a) cat (b) cd (c) pwd (d) ls